Jordan Canonical Form

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of jordan canonical form, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

LA-05 / Linear Algebra / Report v1

I. TARGET AND IMPORT CONTRACT

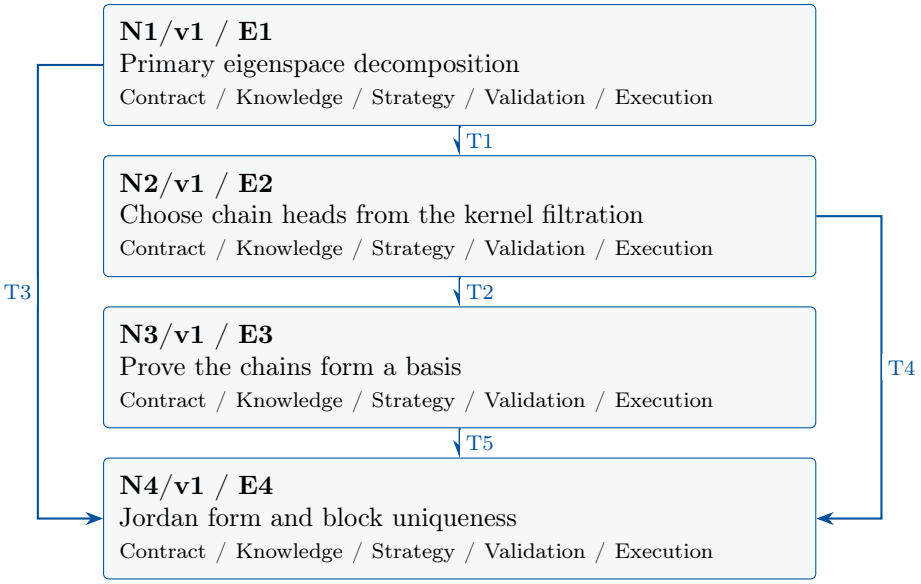
Theorem I.1. *For every complex square matrix A, prove $A = PJP^{-1}$ for a block-diagonal Jordan matrix J, and prove its Jordan blocks are unique up to permutation.*

A. Foundations and permitted imports

Work over C. Import characteristic-polynomial splitting (from the fundamental theorem of algebra), Cayley–Hamilton $\chi_A(A) = 0$, and polynomial Bézout for coprime polynomials. Define $K_j = \ker N^j$ for a nilpotent restriction. The primary decomposition and nilpotent-chain basis are constructed below, so they are not hidden imports.

For each nonzero primary summand let s be the nilpotency index, set $K_0 = 0$, and set $K_{s+1} = K_s = V_\lambda$. Finite-dimensional basis extension and rank–nullity (LA-01) justify complement choices and dimension counts.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. A direct sum of generalized eigenspaces with nilpotent restrictions.

Receiving activity. N2: Choose chain heads from the kernel filtration

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Nilpotent N with $K_s = V$, $K_0 = 0$, $K_{s+1} = V$; prior exports E1/v1

T2: N2/E2/v1 to N3. Chain heads of exact lengths j, with counts w_j.

Receiving activity. N3: Prove the chains form a basis

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Complement identities for every K_j; prior exports E2/v1

T3: N1/E1/v1 to N4. A direct sum of generalized eigenspaces with nilpotent restrictions.

Receiving activity. N4: Jordan form and block uniqueness

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Chain bases in all V_λ; prior exports E1/v1, E2/v1, E3/v1

T4: N2/E2/v1 to N4. Chain heads of exact lengths j, with counts w_j.

Receiving activity. N4: Jordan form and block uniqueness

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Chain bases in all V_λ; prior exports E1/v1, E2/v1, E3/v1

T5: N3/E3/v1 to N4. A basis of nilpotent Jordan chains on each generalized eigenspace.

Receiving activity. N4: Jordan form and block uniqueness

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Chain bases in all V_λ; prior exports E1/v1, E2/v1, E3/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Primary eigenspace decomposition

Object and role. Construct polynomial projection operators using Bézout.

Inputs. $\chi_A(t) = \prod_\lambda (t-\lambda)^{m_\lambda}$; Cayley–Hamilton; foundational imports stated in §1

C — Contract

Own E1/v1: A direct sum of generalized eigenspaces with nilpotent restrictions.

K — Knowledge

Retain $\chi_A(t) = \prod_\lambda (t-\lambda)^{m_\lambda}$; Cayley–Hamilton; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Construct polynomial projection operators using Bézout.

V — Validation

Ordinary eigenspaces alone need not span V.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Set $p_\lambda = (t-\lambda)^{m_\lambda}$, $P = \prod p_\lambda$ and $Q_\lambda = P/p_\lambda$.

N1.2 Since $\gcd(p_\lambda, Q_\lambda) = 1$ choose b_λ with $b_\lambda Q_\lambda \equiv 1 \bmod p_\lambda$.

N1.3 Put $e_\lambda = b_\lambda Q_\lambda$; it is 0 modulo every other p_μ.

N1.4 Thus $\sum e_\lambda \equiv 1 \bmod$ P and $e_\lambda e_\mu \equiv 0 \bmod$ P for $\lambda \neq \mu$.

N1.5 Evaluating at A gives projections onto $V_\lambda = \ker p_\lambda(A)$: $p_\lambda(A)e_\lambda(A) = 0$, and $e_\lambda(A)$ acts as identity on V_λ and zero on the other V_μ.

N1.6 Therefore V is their direct sum.

N1.7 On V_λ, $N = A - \lambda I$ is nilpotent.

Export E1/v1. A direct sum of generalized eigenspaces with nilpotent restrictions.

Acceptance gate. Ordinary eigenspaces alone need not span V. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Ordinary eigenspaces alone need not span V. Recheck dependent nodes N2, N3, N4.

B. N2/v1: Choose chain heads from the kernel filtration

Object and role. Complement the lower layer plus images of longer chains.

Inputs. Nilpotent N with $K_s = V$, $K_0 = 0$, $K_{s+1} = V$; prior exports E1/v1

C — Contract

Own E2/v1: Chain heads of exact lengths j, with counts w_j.

K — Knowledge

Retain Nilpotent N with $K_s = V$, $K_0 = 0$, $K_{s+1} = V$; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Complement the lower layer plus images of longer chains.

V — Validation

An arbitrary basis of kerN cannot serve as chain heads for all lengths.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 For each $j = 1, \ldots, s$ choose a subspace W_j complementary to $K_{j-1} + N\ K_{j+1}$ in K_j.

N2.2 The denominator lies in K_j.

N2.3 Its intersection $K_{j-1} \cap N\ K_{j+1}$ equals N K_j: if Nz lies in K_{j-1}, $N^j z = 0$ so $z \in K_j$.

N2.4 Writing $d_j = \dim K_j$ and $d_0 = 0, d_{s+1} = d_s$, and using dimN $K_j = d_j - d_1$, gives $w_j = \dim W_j = 2d_j - d_{j-1} - d_{j+1}$.

N2.5 A nonzero chosen basis vector in W_j does not lie in K_{j-1}, so its N-chain has exact lengthj.

Export E2/v1. Chain heads of exact lengths j, with counts w_j.

Acceptance gate. An arbitrary basis of kerN cannot serve as chain heads for all lengths. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: An arbitrary basis of kerN cannot serve as chain heads for all lengths. Recheck dependent nodes N3, N4.

C. N3/v1: Prove the chains form a basis

Object and role. Show spanning and count exactly dimV vectors.

Inputs. Complement identities for every K_j; prior exports E2/v1

C — Contract

Own E3/v1: A basis of nilpotent Jordan chains on each generalized eigenspace.

K — Knowledge

Retain Complement identities for every K_j; prior exports E2/v1; keep their assumptions and versions attached.

S — Strategy

Show spanning and count exactly dimV vectors.

V — Validation

Both spanning and exact count are needed; a collection of chains may otherwise be dependent.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 The identity $K_j = K_{j-1} + W_j + N\ K_{j+1}$ can be expanded upwards.

N3.2 Terms N K_j lie in K_{j-1}, giving $K_j \subset K_{j-1} + \sum_{\ell=j}^{s} N^{\ell-j} W_\ell$.

N3.3 Descend through $j = s, \ldots, 1$; this shows all of V is spanned by the chains $w, Nw, \ldots, N^{j-1}w$ for basis heads $w \in W_j$.

N3.4 Their total number is $\sum_{j=1}^{s}$ j $w_j = \sum j(2d_j - d_{j-1} - d_{j+1}) = d_s$ by telescoping.

N3.5 A spanning list of exactly dimV vectors is a basis.

N3.6 Order each chain $as(N^{j-1}w, \ldots, Nw, w)$; N then has ones on the superdiagonal and zeros elsewhere in that block.

Export E3/v1. A basis of nilpotent Jordan chains on each generalized eigenspace.

Acceptance gate. Both spanning and exact count are needed; a collection of chains may otherwise be dependent. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Both spanning and exact count are needed; a collection of chains may otherwise be dependent. Recheck dependent nodes N4.

D. N4/v1: Jordan form and block uniqueness

Object and role. Read block multiplicities from invariant kernel dimensions.

Inputs. Chain bases in all V_λ; prior exports E1/v1, E2/v1, E3/v1

C — Contract

Own E4/v1: Existence and uniqueness up to permutation of complex Jordan blocks.

K — Knowledge

Retain Chain bases in all V_λ; prior exports E1/v1, E2/v1, E3/v1; keep their assumptions and versions attached.

S — Strategy

Read block multiplicities from invariant kernel dimensions.

V — Validation

Uniqueness concerns block data, not the chain basis or change-of-basis matrix.

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 Joining the bases over distinct λ gives J with blocks $J_j(\lambda)$, so $A = PJP^{-1}$ for the change-of-basis matrix P.

N4.2 A length ℓ block contributes $\min(j, \ell)$ to $\dim \ker N^j$.

N4.3 Thus $d_j - d_{j-1}$ is the number of blocks of length at leastj, and $w_j = (d_j - d_{j-1}) - (d_{j+1} - d_j)$ is the number of length exactlyj.

N4.4 These kernel dimensions are invariant under similarity, as are the eigenvalues.

N4.5 Hence every block multiplicity is determined, proving uniqueness up to block order.

Export E4/v1. Existence and uniqueness up to permutation of complex Jordan blocks.

Acceptance gate. Uniqueness concerns block data, not the chain basis or change-of-basis matrix. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Uniqueness concerns block data, not the chain basis or change-of-basis matrix. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Primary eigenspace decomposition

1. Set $p_\lambda = (t-\lambda)^{m_\lambda}$, $P = \prod p_\lambda$ and $Q_\lambda = P/p_\lambda$.
2. Since $\gcd(p_\lambda, Q_\lambda) = 1$ choose b_λ with $b_\lambda Q_\lambda \equiv 1 \bmod p_\lambda$.
3. Put $e_\lambda = b_\lambda Q_\lambda$; it is 0 modulo every other p_μ.
4. Thus $\sum e_\lambda \equiv 1$ mod P and $e_\lambda e_\mu \equiv 0$ mod P for $\lambda \neq \mu$.
5. Evaluating at A gives projections onto $V_\lambda = \ker p_\lambda(A)$: $p_\lambda(A)e_\lambda(A) = 0$, and $e_\lambda(A)$ acts as identity on V_λ and zero on the other V_μ.
6. Therefore V is their direct sum.
7. On V_λ, $N = A - \lambda I$ is nilpotent.

B. §4.N2 — Choose chain heads from the kernel filtration

On a primary summand let s be the nilpotency index, $K_j = \ker N^j$, and $K_{s+1} = K_s$.

1. For each $j = 1, \ldots, s$ choose a subspace W_j complementary to $K_{j-1} + N\ K_{j+1}$ in K_j.
2. The denominator lies in K_j.
3. Its intersection $K_{j-1} \cap N\ K_{j+1}$ equals N K_j: if Nz lies in K_{j-1}, $N^j z = 0$ so $z \in K_j$.
4. Writing $d_j = \dim K_j$ and $d_0 = 0, d_{s+1} = d_s$, and using dimN $K_j = d_j - d_1$, gives $w_j = \dim W_j = 2d_j - d_{j-1} - d_{j+1}$.
5. A nonzero chosen basis vector in W_j does not lie in K_{j-1}, so its N-chain has exact lengthj.

C. §4.N3 — Prove the chains form a basis

1. The identity $K_j = K_{j-1} + W_j + N\ K_{j+1}$ can be expanded upwards.
2. Terms N K_j lie in K_{j-1}, giving $K_j \subset K_{j-1} + \sum_{\ell=j}^{s} N^{\ell-j} W_\ell$.
3. Descend through $j = s, \ldots, 1$; this shows all of V is spanned by the chains $w, Nw, \ldots, N^{j-1}w$ for basis heads $w \in W_j$.
4. Their total number is $\sum_{j=1}^{s}$ j $w_j = \sum j(2d_j - d_{j-1} - d_{j+1}) = d_s$ by telescoping.

5. A spanning list of exactly dimV vectors is a basis.

6. Order each chain $as(N^{j-1}w, \ldots, Nw, w)$; N then has ones on the superdiagonal and zeros elsewhere in that block.

D. §4.N4 — Jordan form and block uniqueness

1. Joining the bases over distinct λ gives J with blocks $J_j(\lambda)$, so $A = PJP^{-1}$ for the change-of-basis matrix P.

2. A length ℓ block contributes $\min(j, \ell)$ to $\dim \ker N^j$.

3. Thus $d_j - d_{j-1}$ is the number of blocks of length at leastj, and $w_j = (d_j - d_{j-1}) - (d_{j+1} - d_j)$ is the number of length exactlyj.

4. These kernel dimensions are invariant under similarity, as are the eigenvalues.

5. Hence every block multiplicity is determined, proving uniqueness up to block order.

□

V. WORKED ORCHESTRATION

For a nilpotent operator with kernel dimensions $d_1 = 2, d_2 = 3, d_3 = 4$ and $d_4 = 4$, the increments are 2,1,1,0. There are two blocks total, one of length 3 and one of length 1. Thus $J = J_3(0) \oplus J_1(0)$, and no block of length 2 occurs.

A. Boundary and hypothesis audit

Over R a rotation matrix has no real eigenvalue, so a real Jordan matrix with real diagonal entries cannot represent it. Validation keeps the complex-field contract; a real canonical form needs *separate*2×2 blocks for conjugate eigenvalues.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.

2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

Over R a rotation matrix has no real eigenvalue, so a real Jordan matrix with real diagonal entries cannot represent it. Validation keeps the complex-field contract; a real canonical form needs $separate 2 \times 2$ blocks for conjugate eigenvalues.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- MIT Algebra II notes: field extensions and Galois theory.